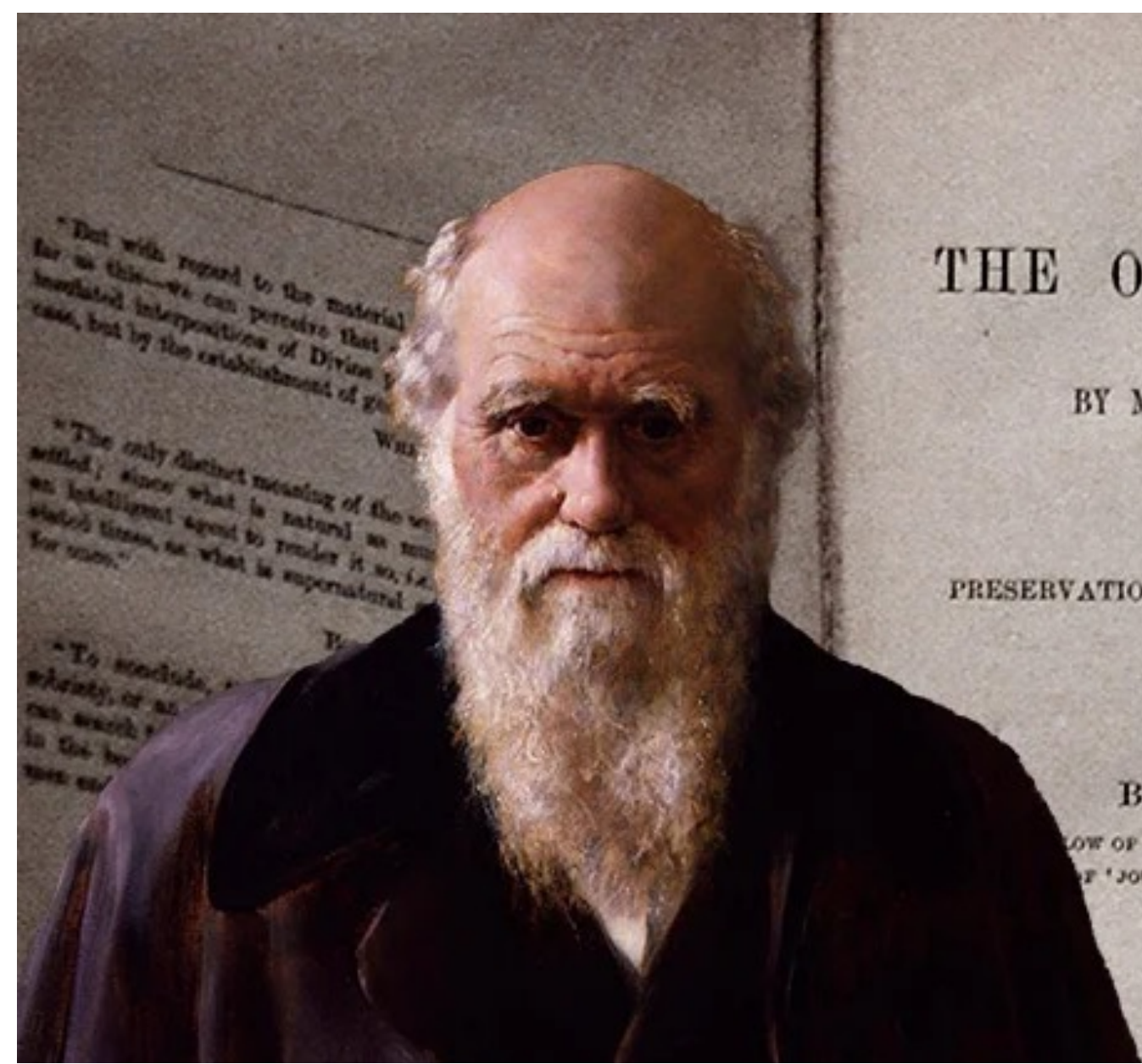




ON
THE ORIGIN OF SPECIES
BY MEANS OF NATURAL SELECTION,
OR THE
PRESERVATION OF FAVOURED RACES IN THE STRUGGLE
FOR LIFE.

By CHARLES DARWIN, M.A.,

FELLOW OF THE ROYAL, GEOLOGICAL, LINNEAN, ETC., SOCIETIES;
AUTHOR OF 'JOURNAL OF RESEARCHES DURING H. M. S. BEAGLE'S VOYAGE
ROUND THE WORLD.'

FIFTH THOUSAND.

“A Patchwork Darwinge”

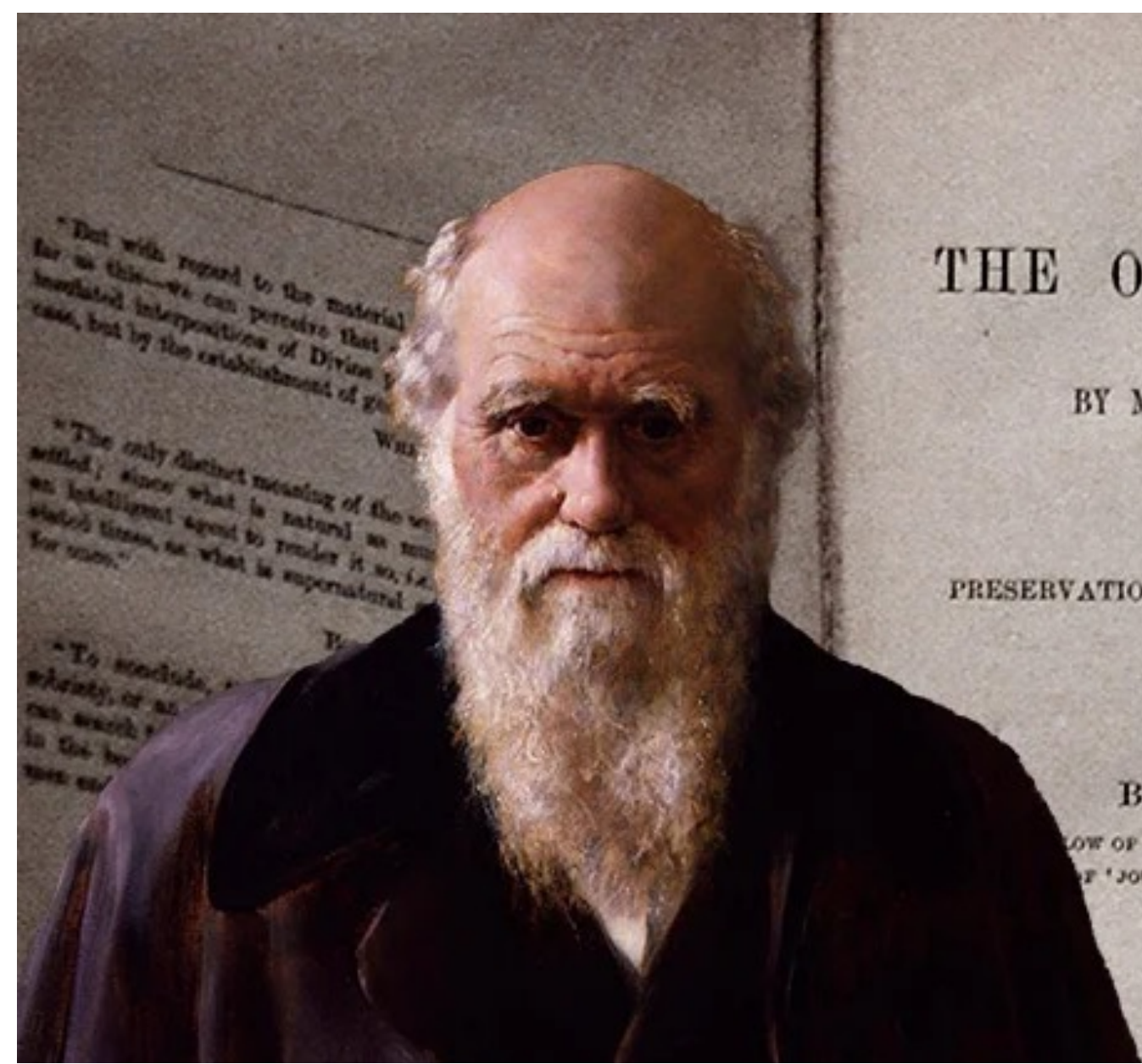
Evolution: Variation and Selection

An illustration of the mechanisms involved in evolutionary processes

18.12.2025

Jacinta Jonetzki

Introduction to Complex Systems



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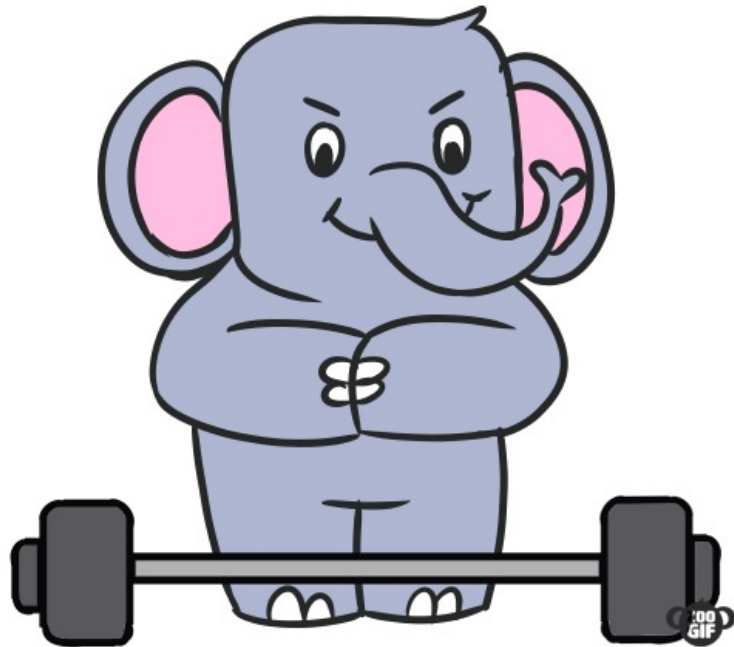
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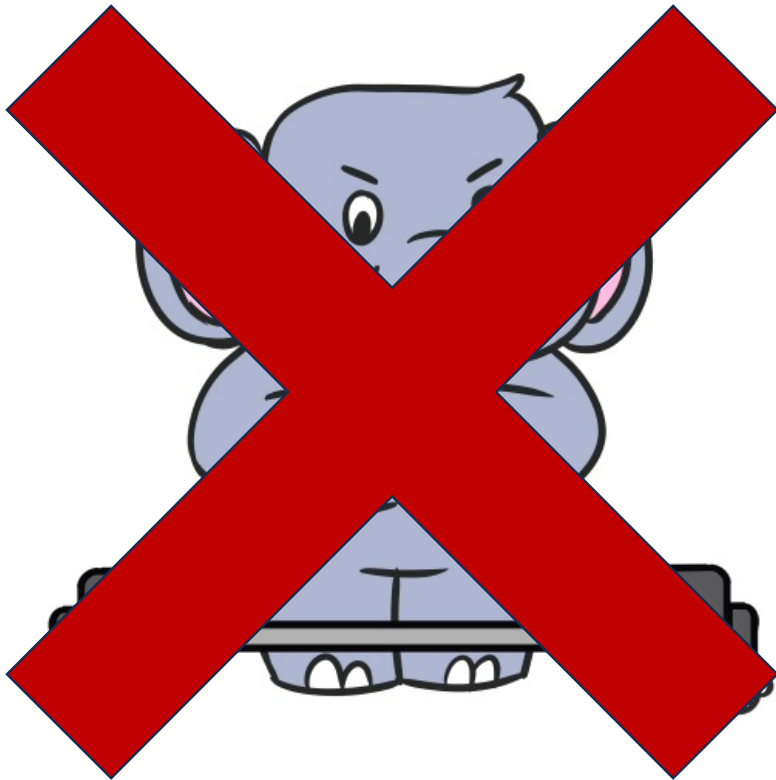
Evolution = Change in the genetic composition of a population over time

Fitness



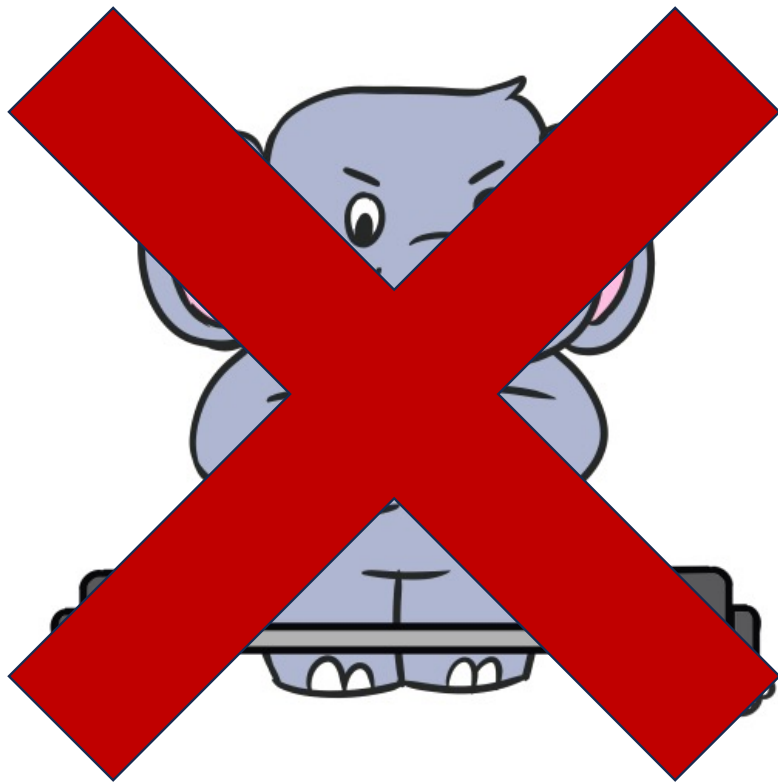
Survival of the fittest

Fitness



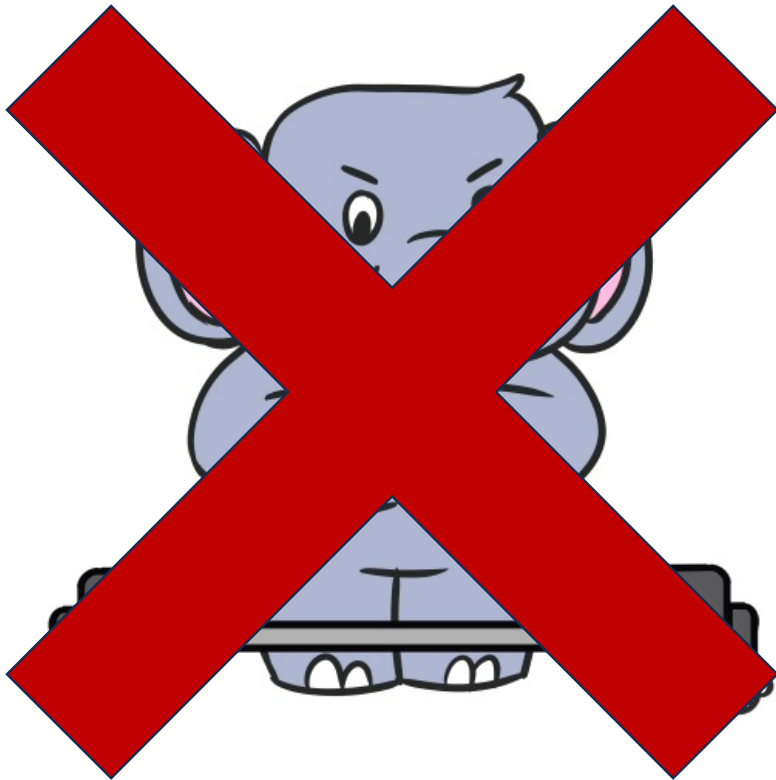
Survival of the fittest

Survival of the fittest



Fitness = Proportion of copies of a gene that an individual leaves behind in the gene pool of the next generation relative to other individuals

Survival of the fittest



Fitness = lifetime
reproductive success →
number of offspring

Variation

- Differences in genetic traits within and between populations

Variation

- Differences in genetic traits within and between populations

(natural) Selection

- Individuals with advantageous traits have a higher reproductive success

Adaptive evolution requires...

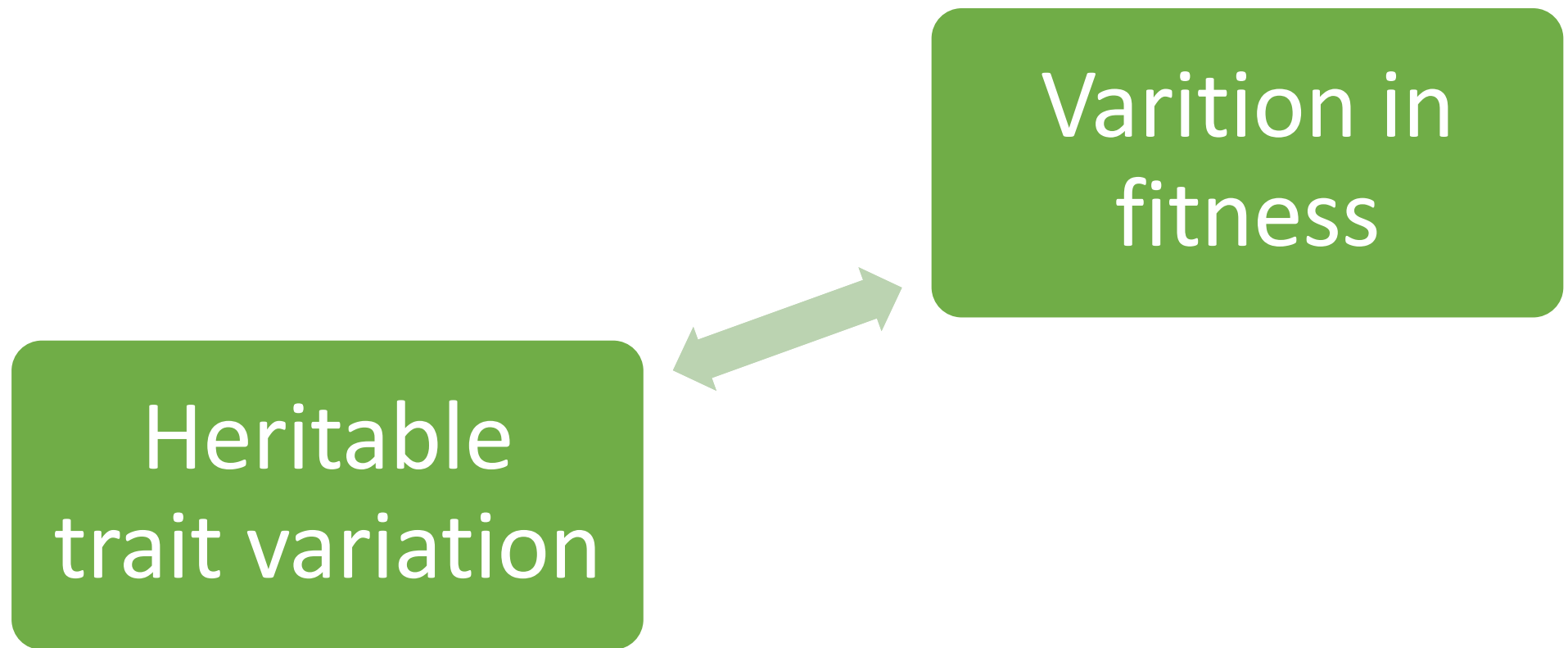
Heritable
trait variation

Adaptive evolution requires...

Variation in
fitness

Heritable
trait variation

Adaptive evolution requires...



“A Patchwork Darwinge”

Evolution: Variation and Selection

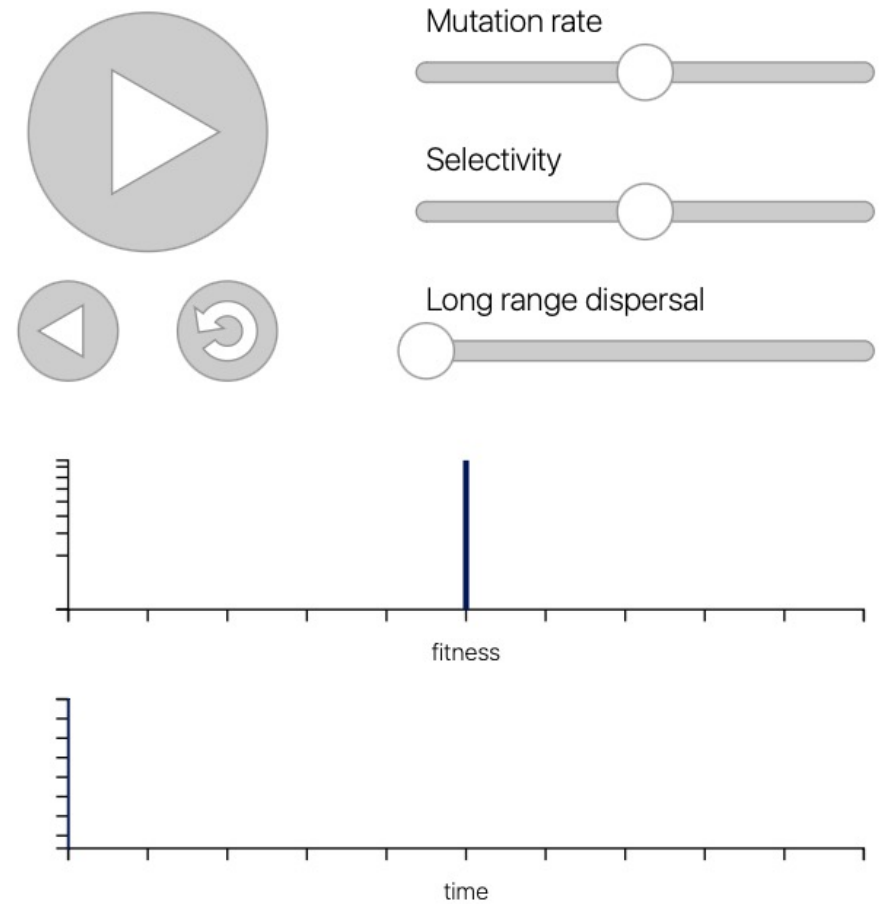
An illustration of the mechanisms involved in evolutionary processes

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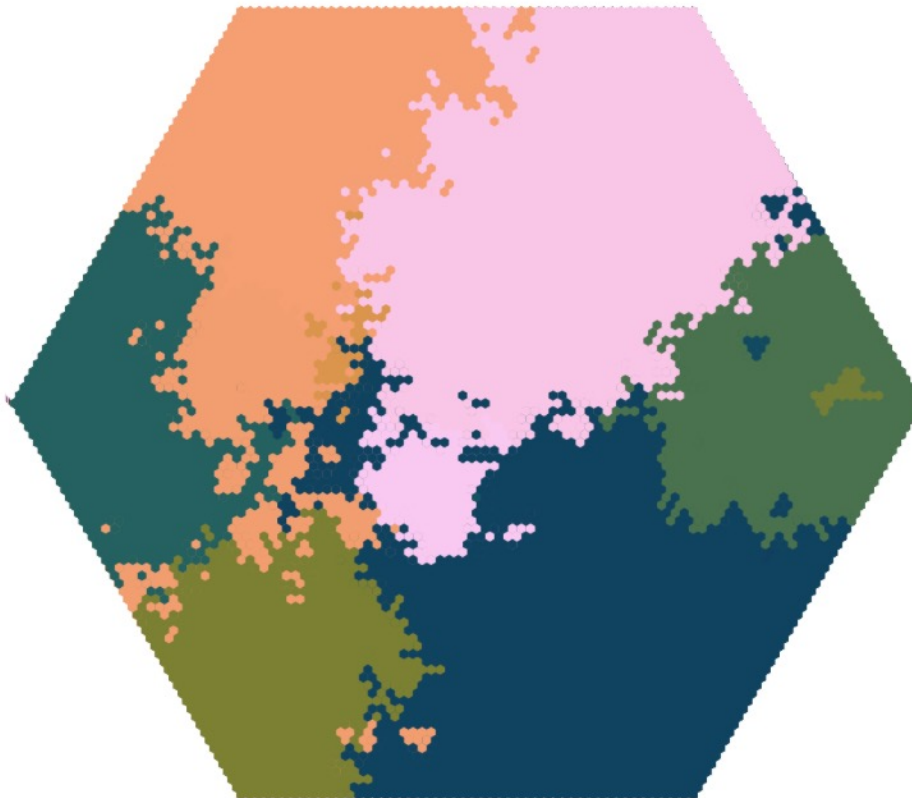
Introduction to Complex Systems

Population of
mutants
(uniform)



<https://www.complexity-explorables.org/explorables/a-patchwork-darwinge/>

Characteristics of evolution in the explorable



Characteristics of evolution in the explorable

**Variation of
traits**



every color
stands for one
mutant type



Characteristics of evolution in the explorable

Variation of traits



every color
stands for one
mutant type



Variation in fitness

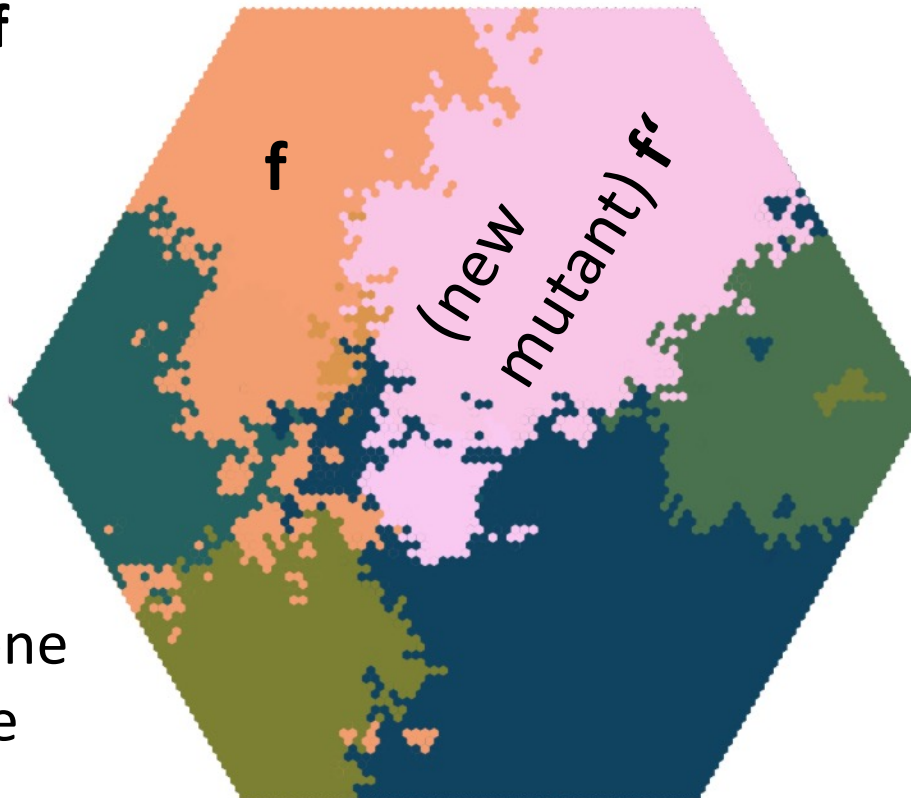
→ every mutant has a specific fitness

Characteristics of evolution in the explorable

Variation of traits



every color stands for one mutant type



Variation in fitness

→ every mutant has a specific fitness



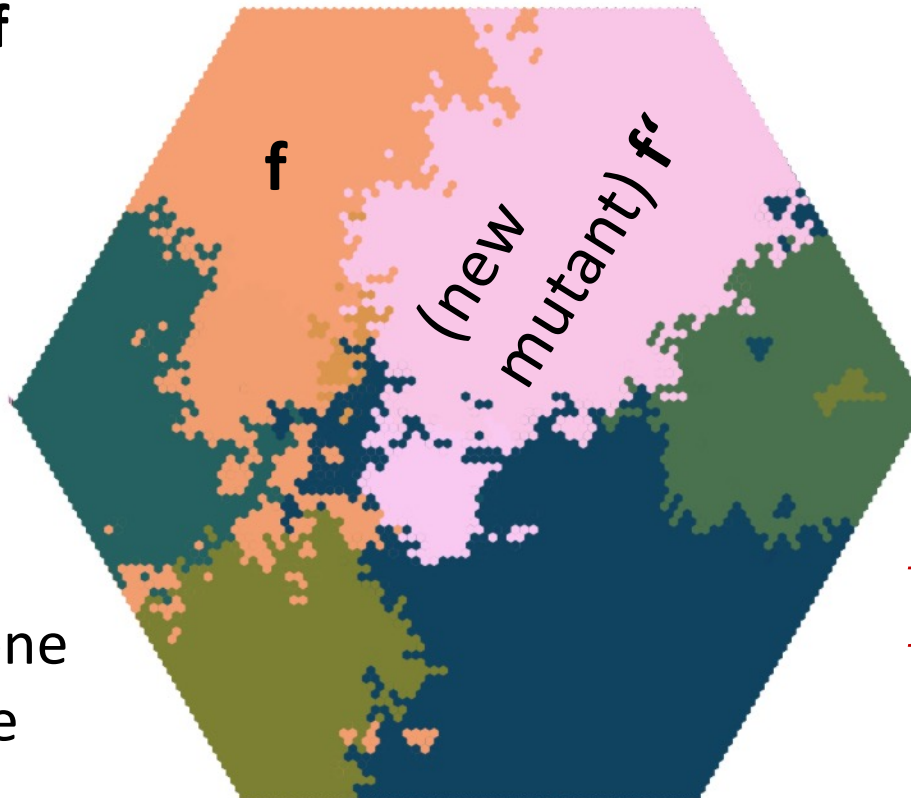
$$f' = f + \delta f$$

Characteristics of evolution in the explorable

Variation of traits



every color stands for one mutant type



Variation in fitness

→ every mutant has a specific fitness

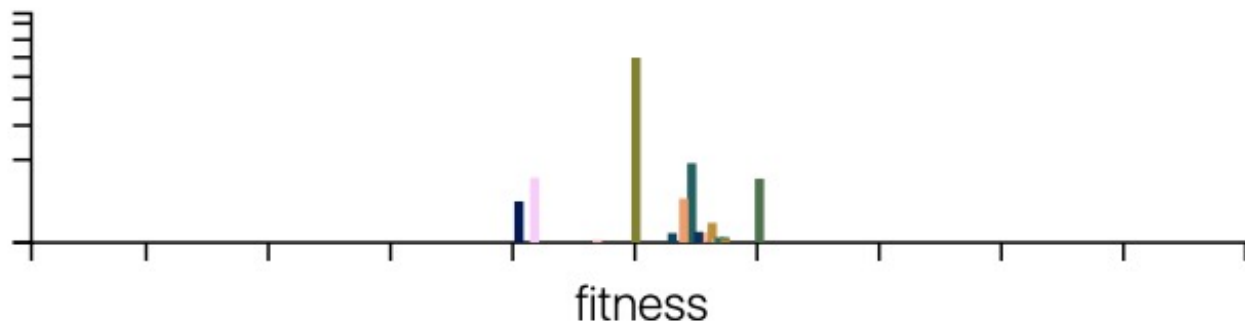


$$f' = f + \delta f$$

$f' < f \rightarrow$ new mutant goes extinct

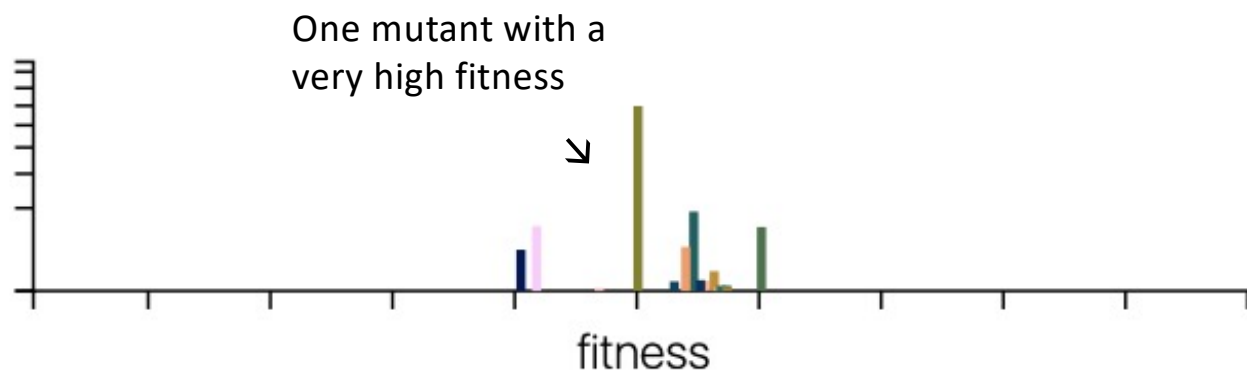
$f' > f \rightarrow$ new mutants dominates

Plots



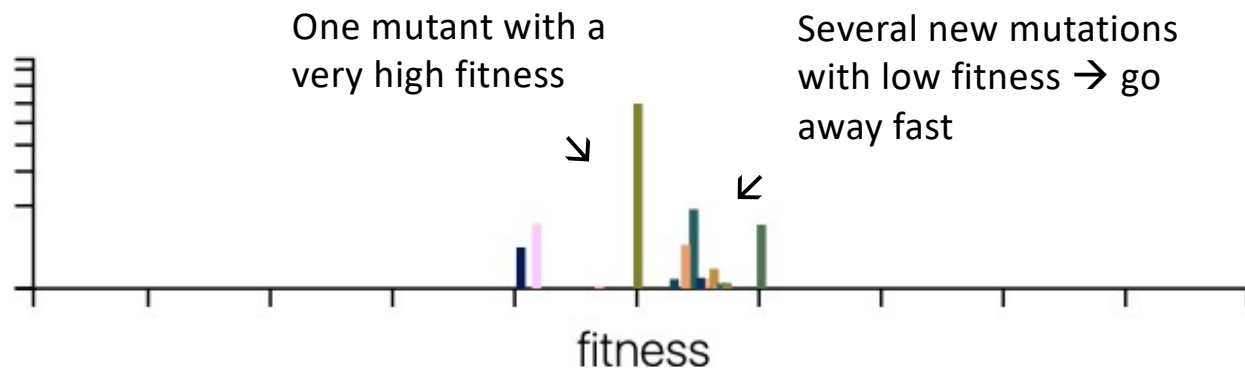
Fitness of current
mutants

Plots



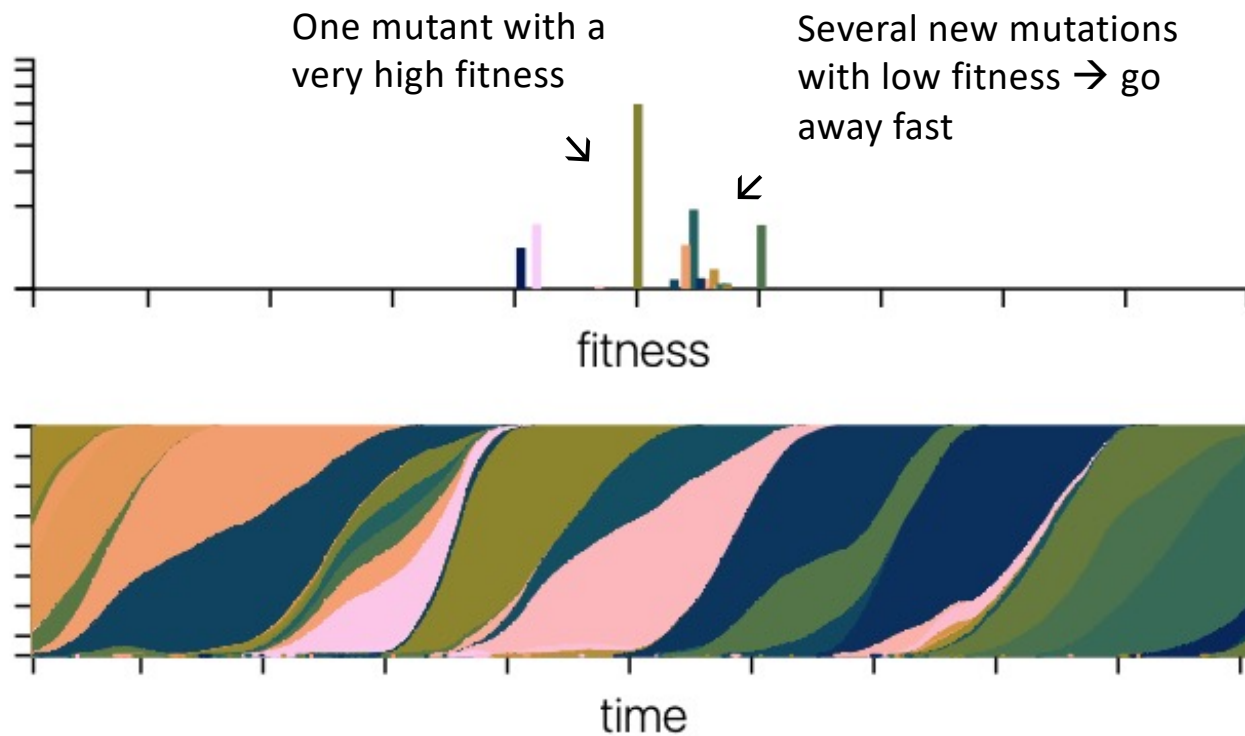
Fitness of current mutants

Plots



Fitness of current mutants

Plots

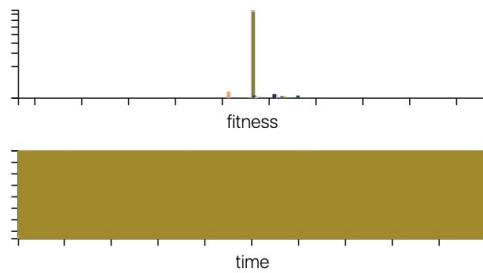
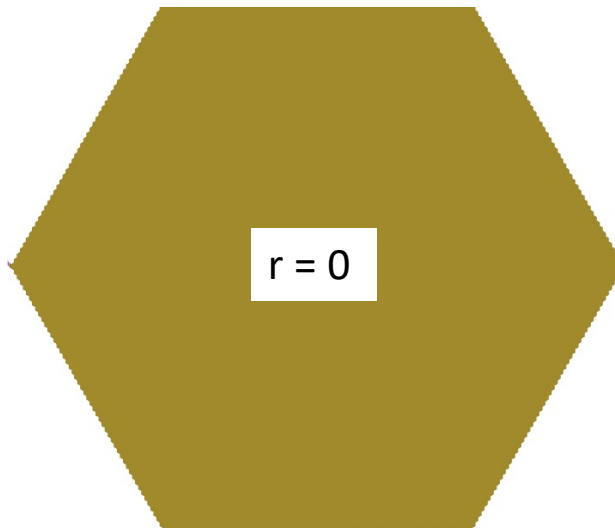


Fitness of current mutants

Abundance of mutants in population

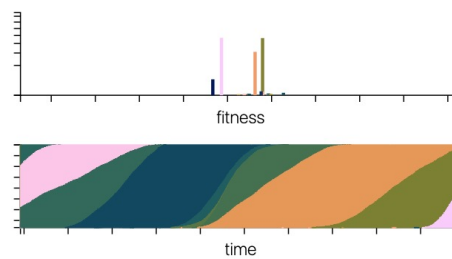
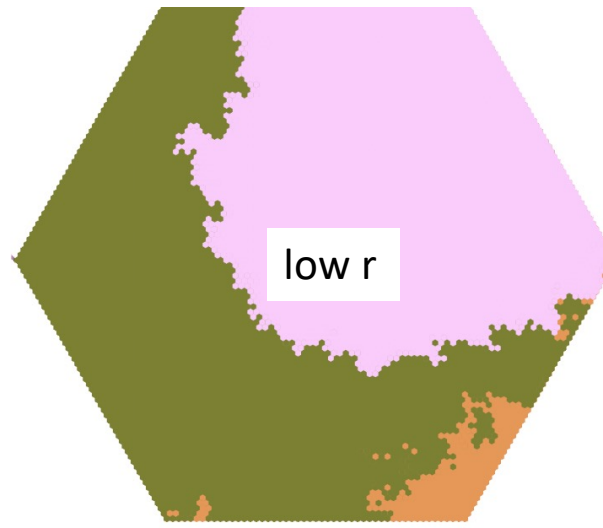
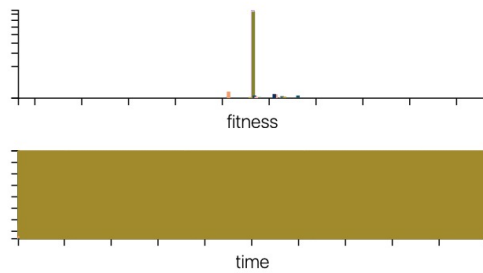
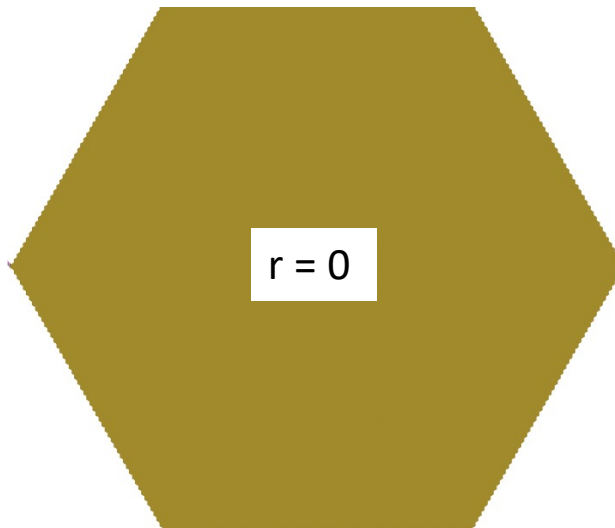
Parameters: Mutation rate (r)

= rate at which mutations occur in system



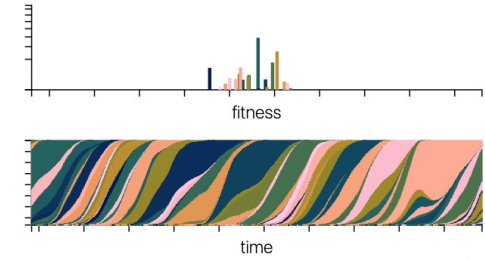
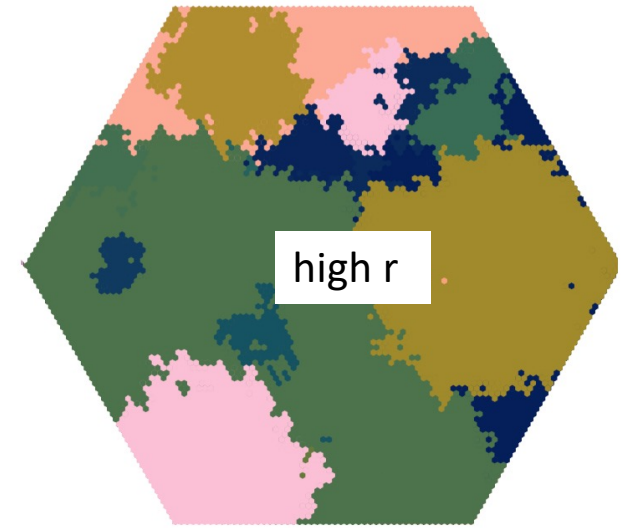
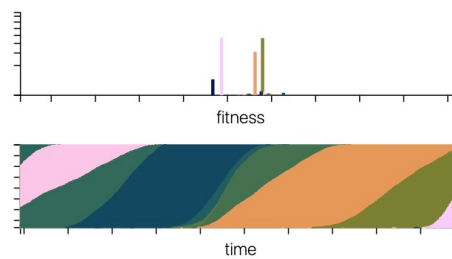
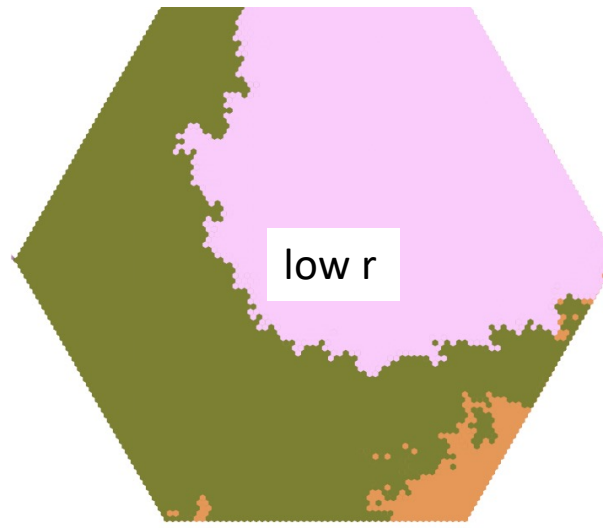
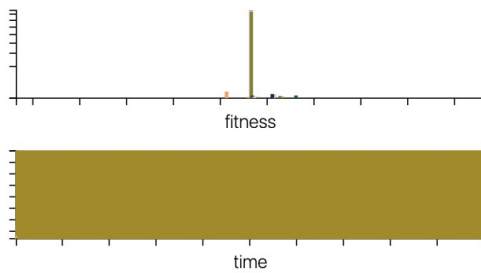
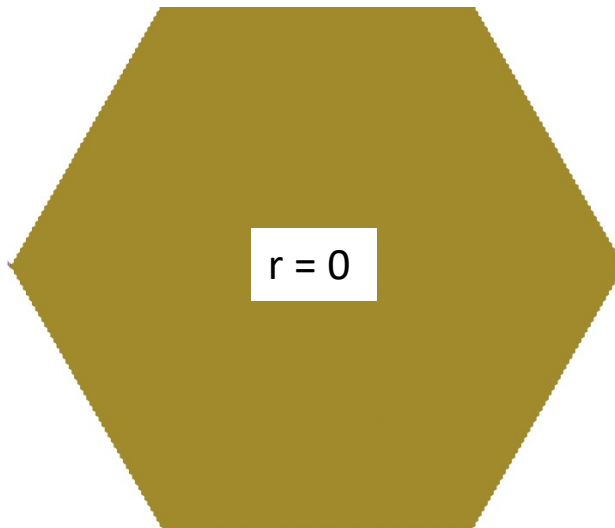
Parameters: Mutation rate (r)

= rate at which mutations occur in system



Parameters: Mutation rate (r)

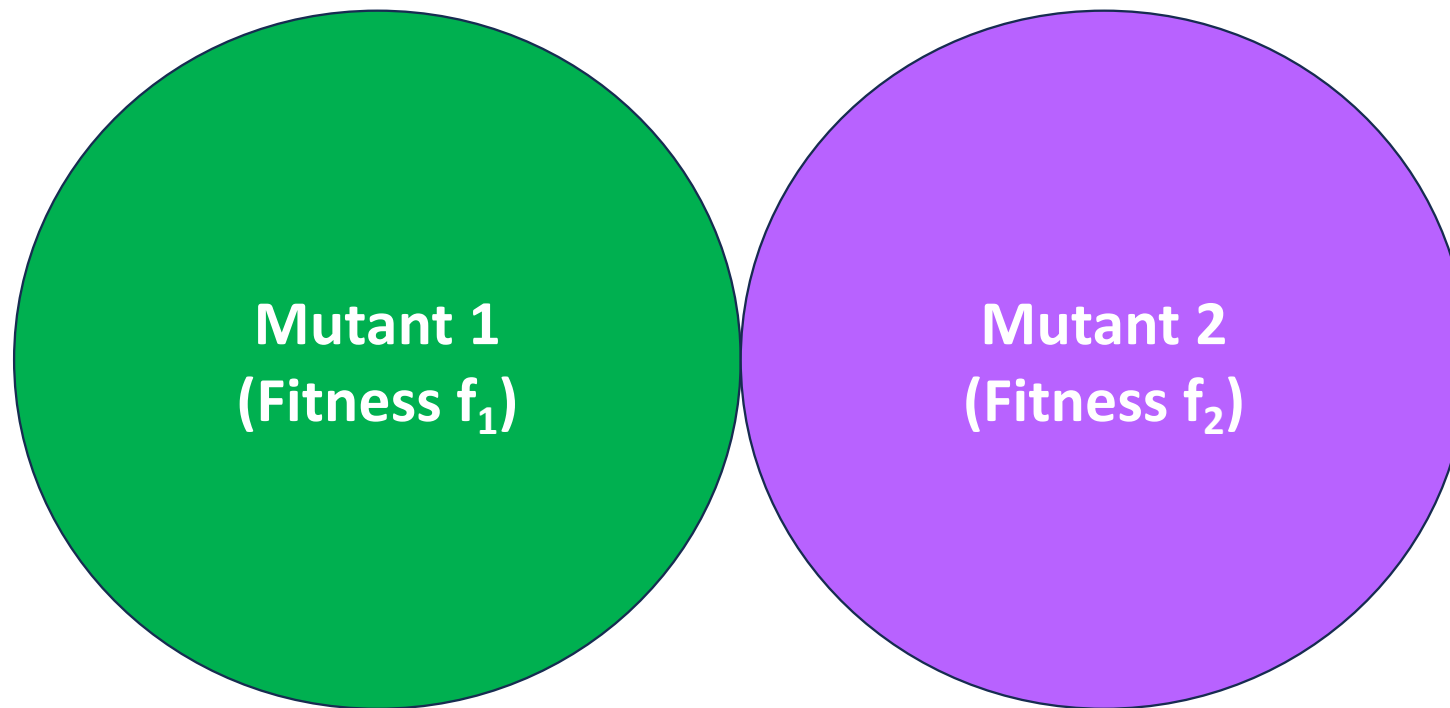
= rate at which mutations occur in system



Parameters: Selectivity

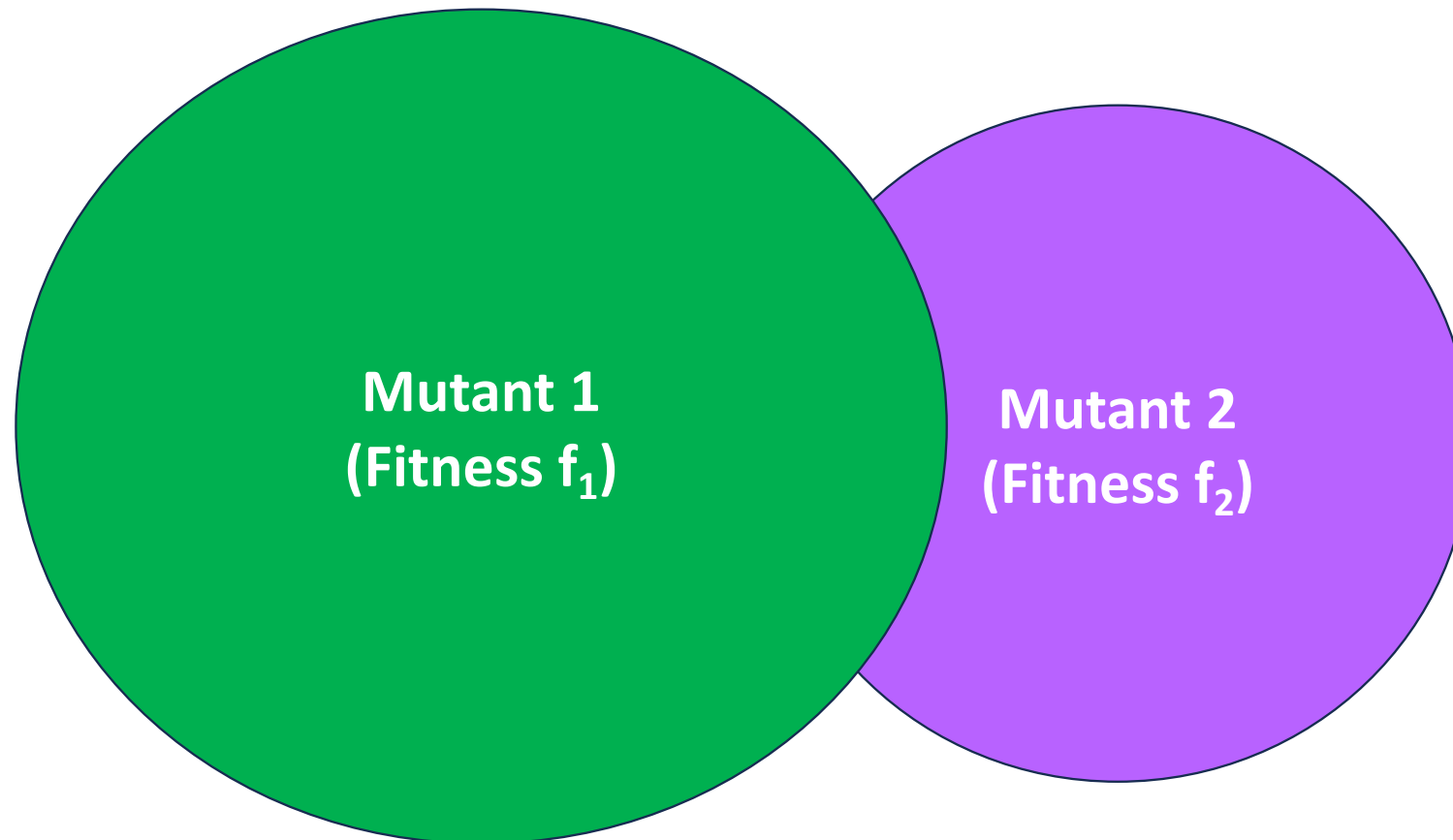


Parameters: Selectivity



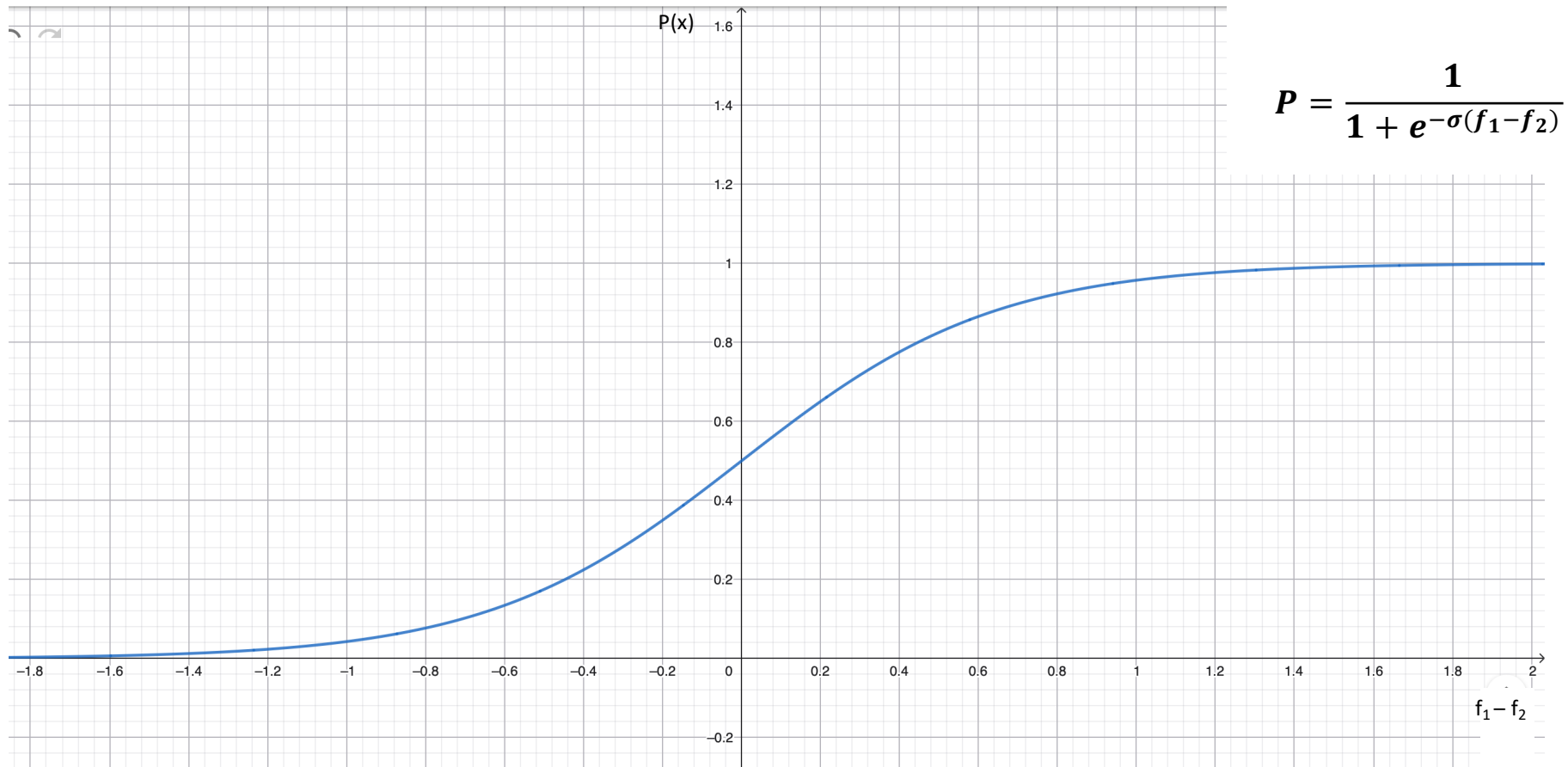
mutant 1 will remove mutant 2 with probability $0 \leq P \leq 1$

Parameters: Selectivity

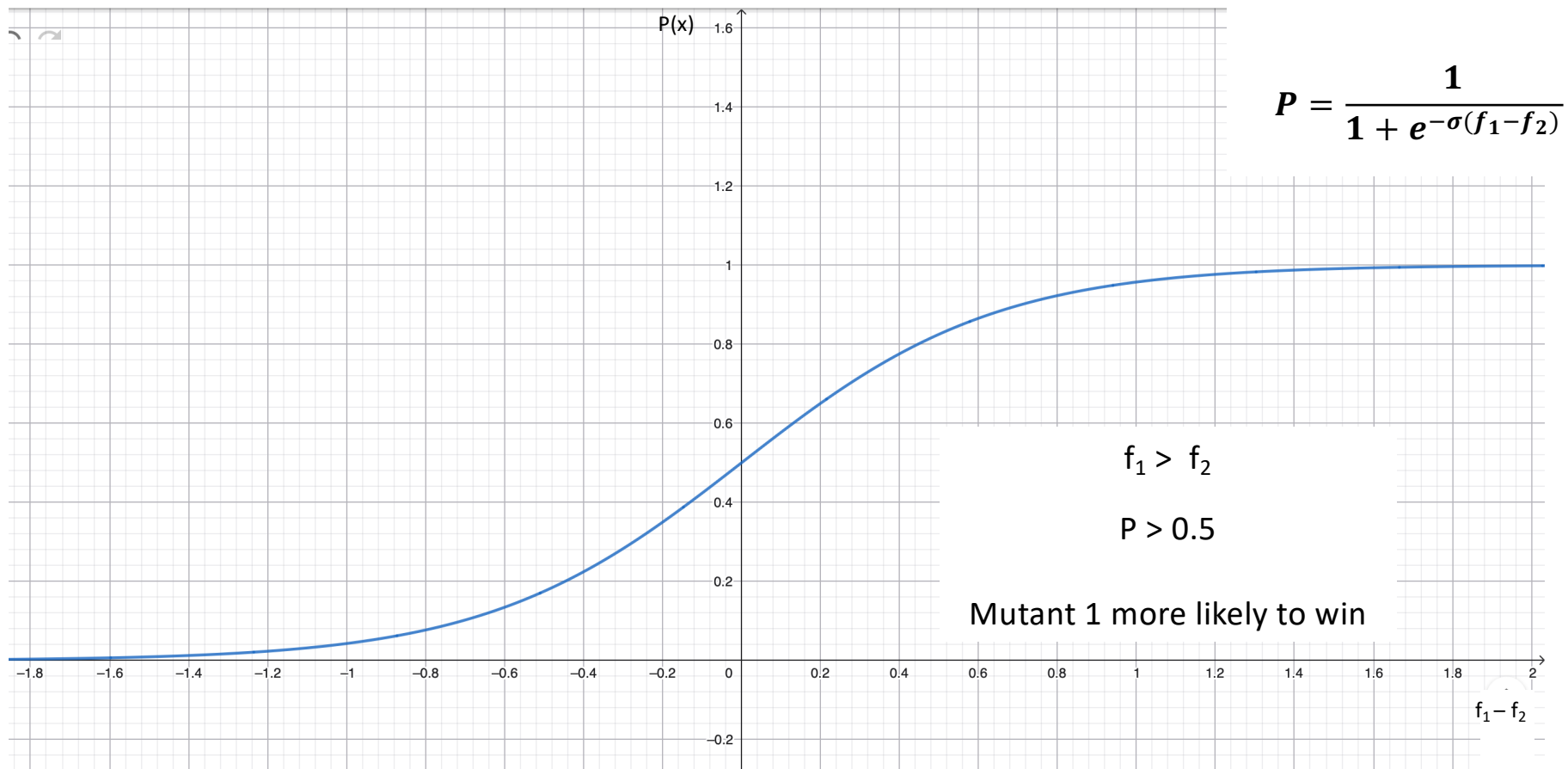


mutant 1 will remove mutant 2 with probability $0 \leq P \leq 1$

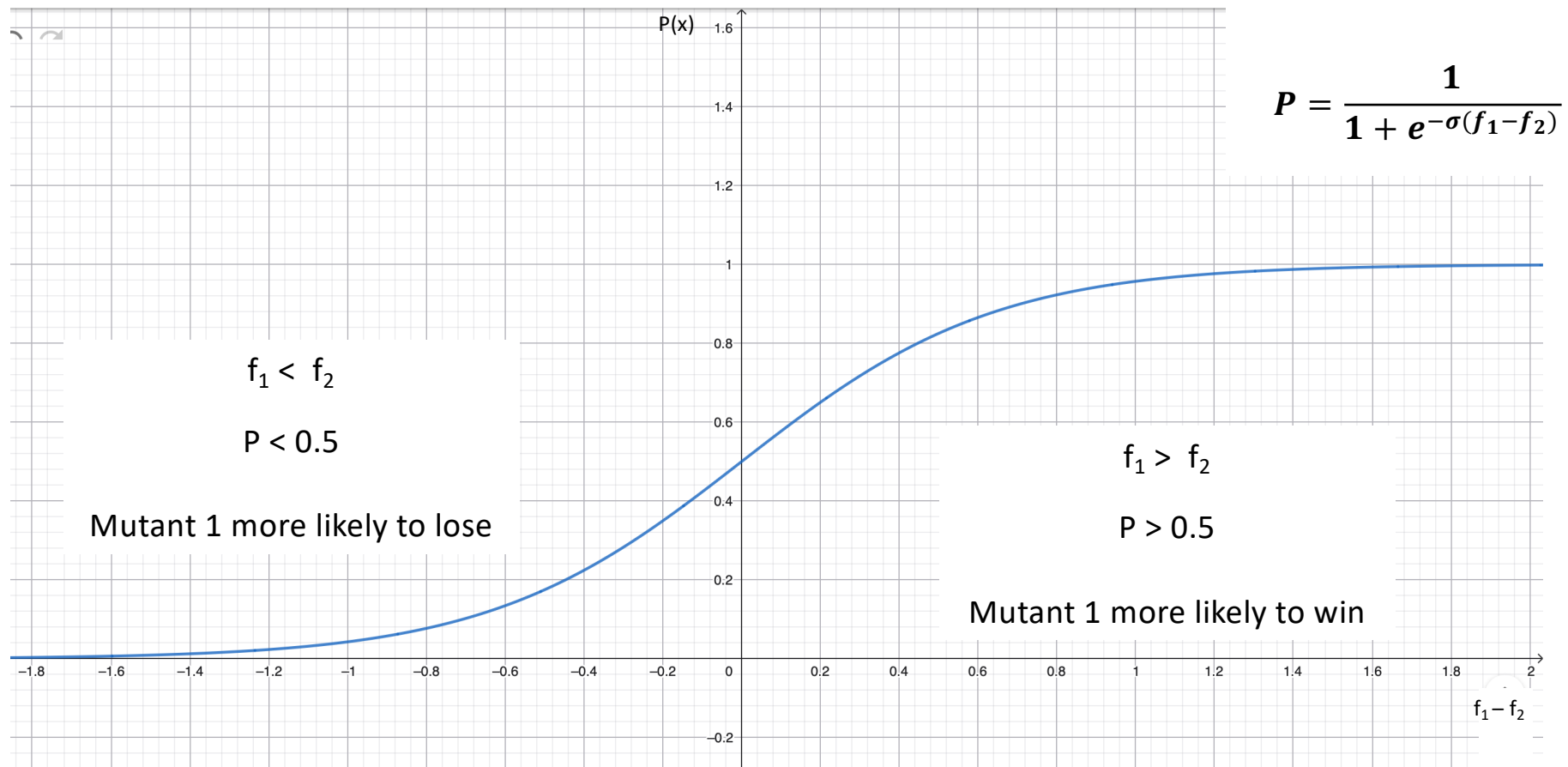
Probability function



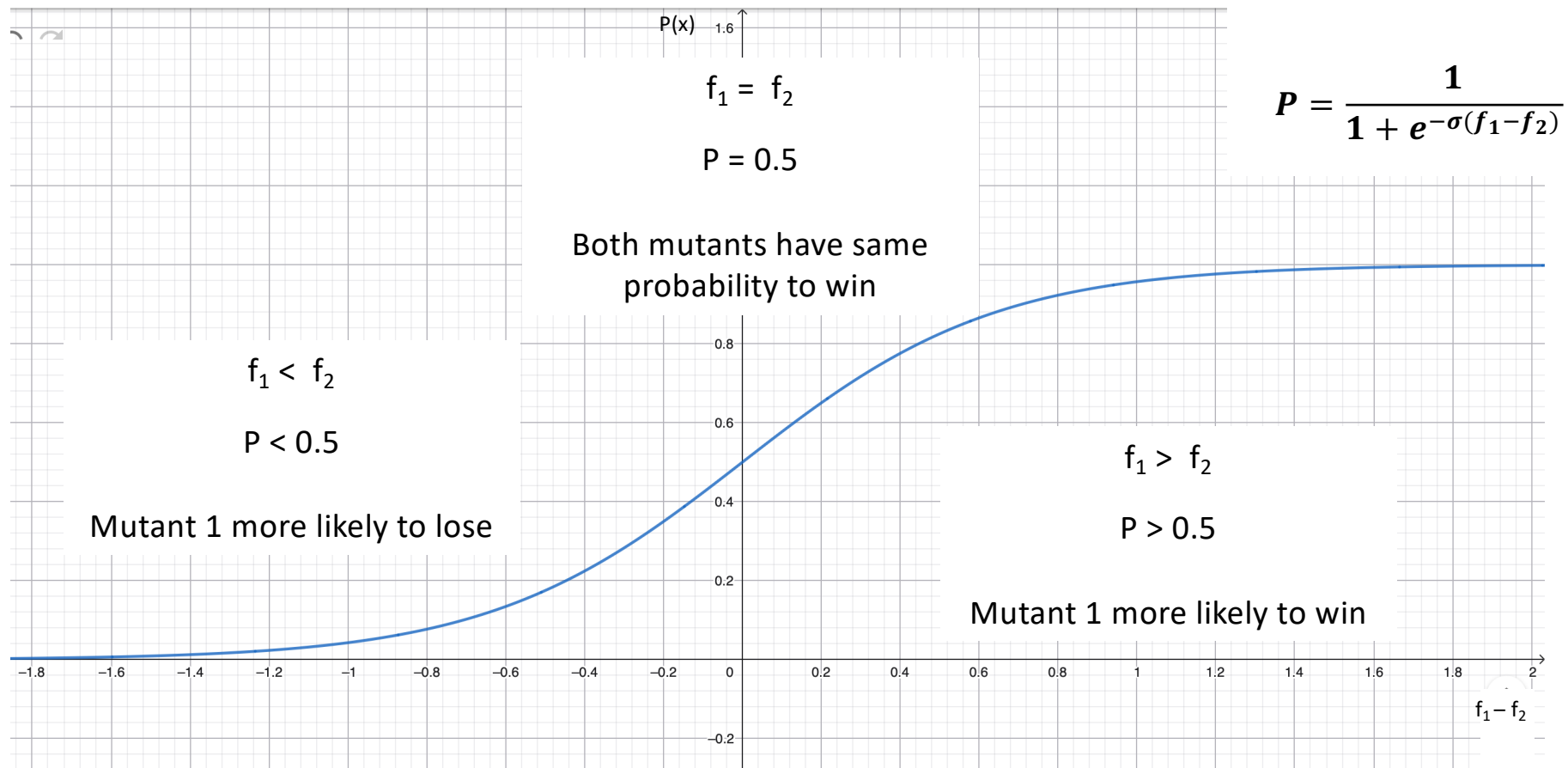
Probability function



Probability function

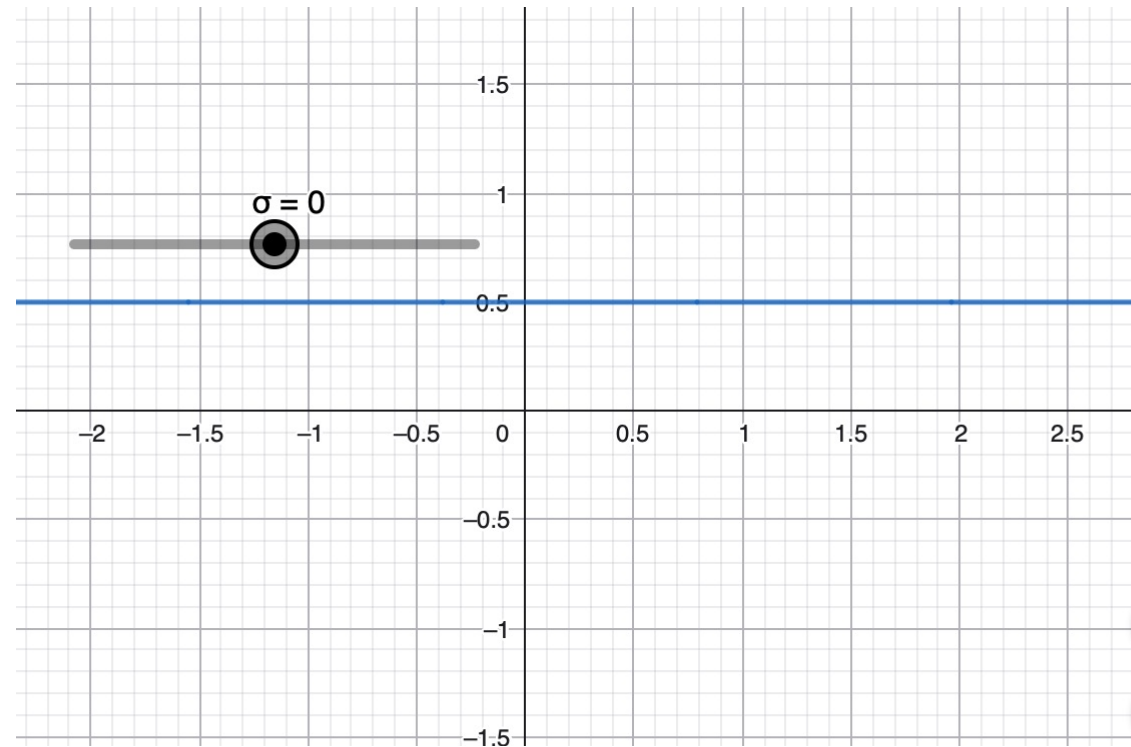
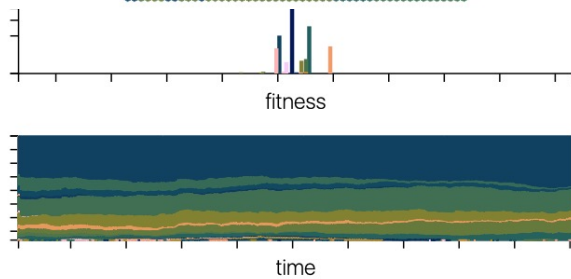
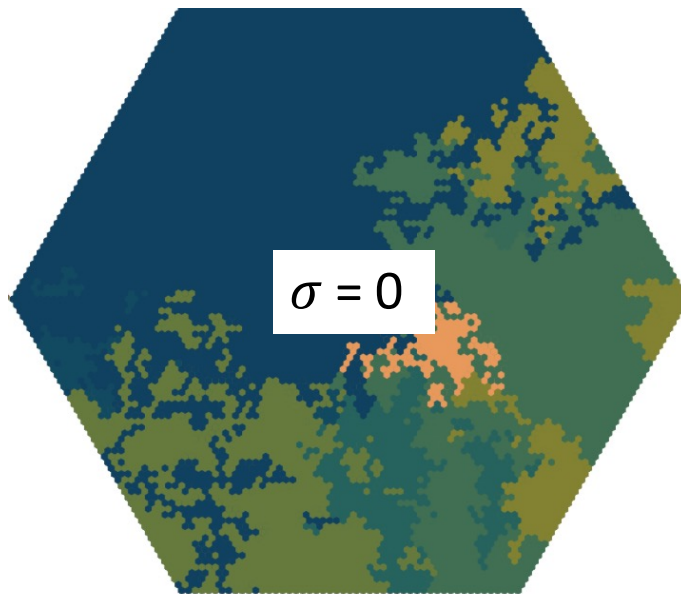


Probability function



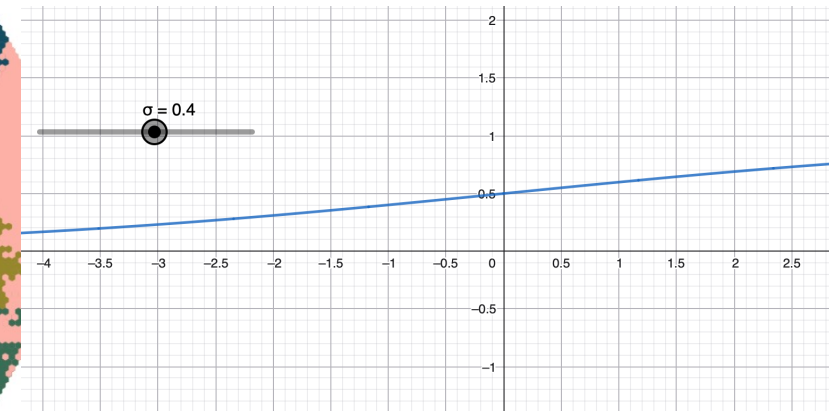
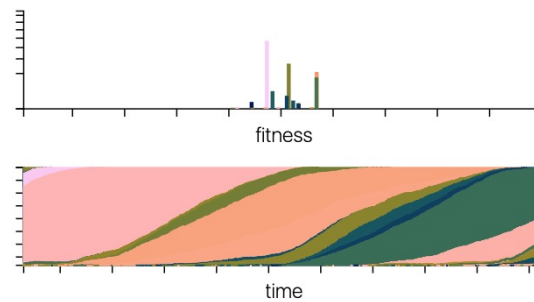
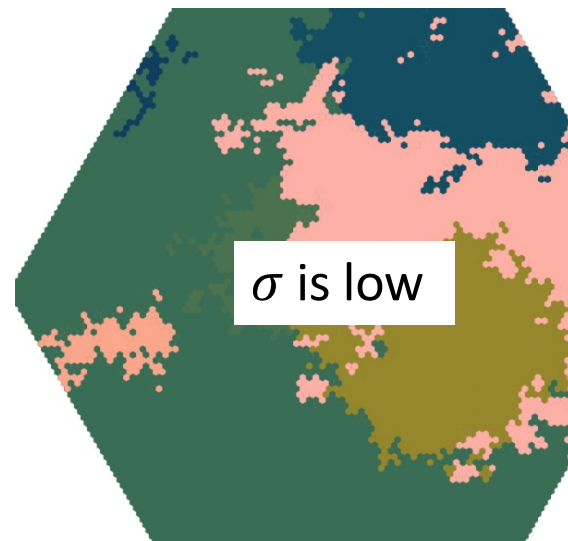
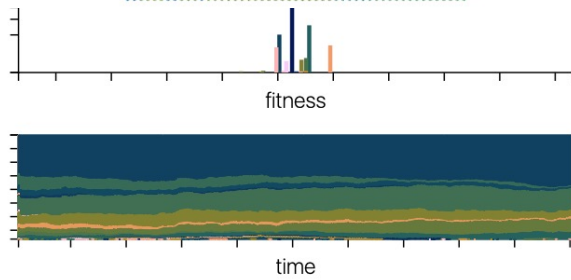
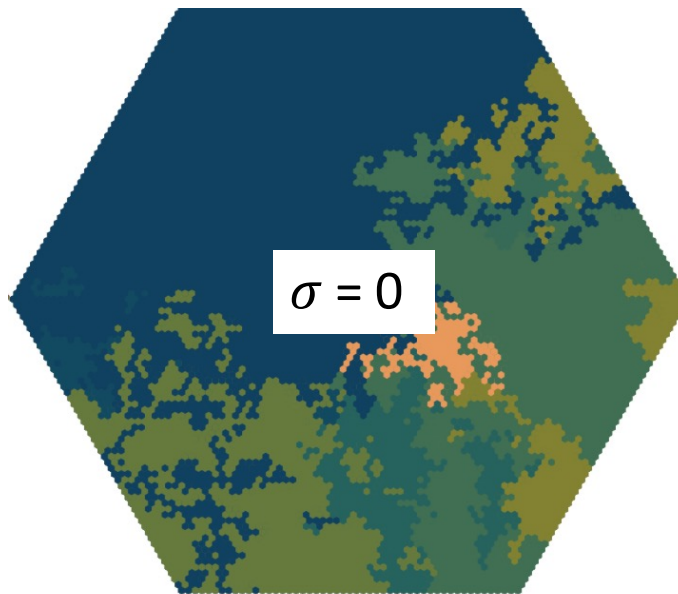
Parameters: Selectivity σ

= controls how clear decision about winner is



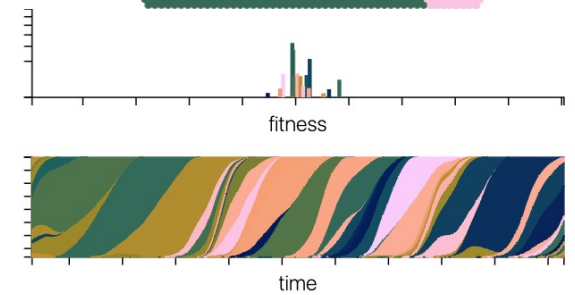
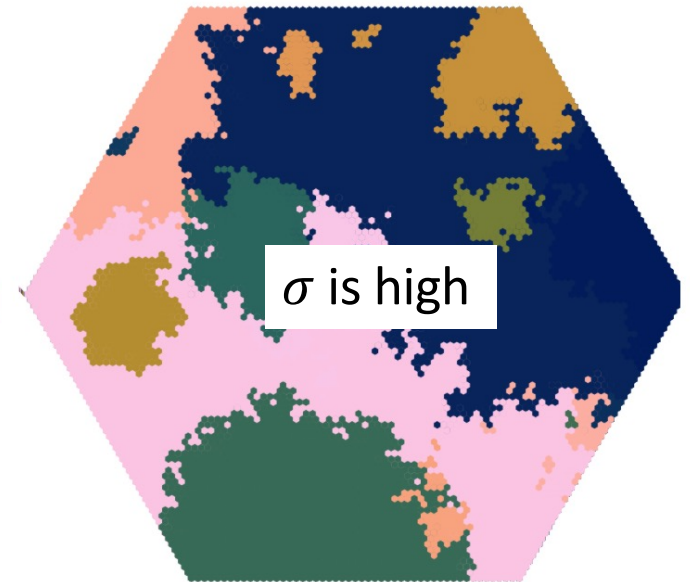
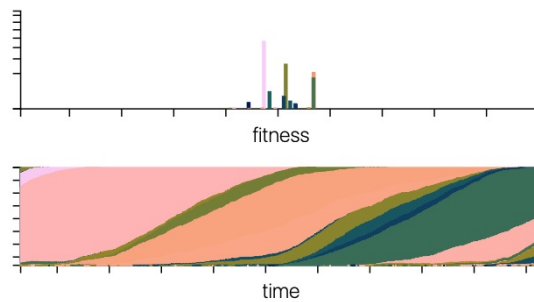
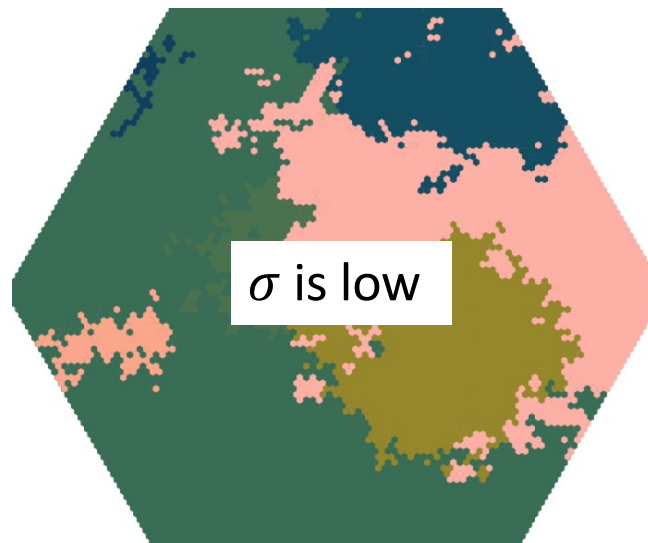
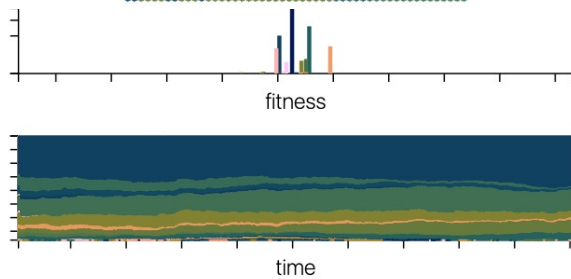
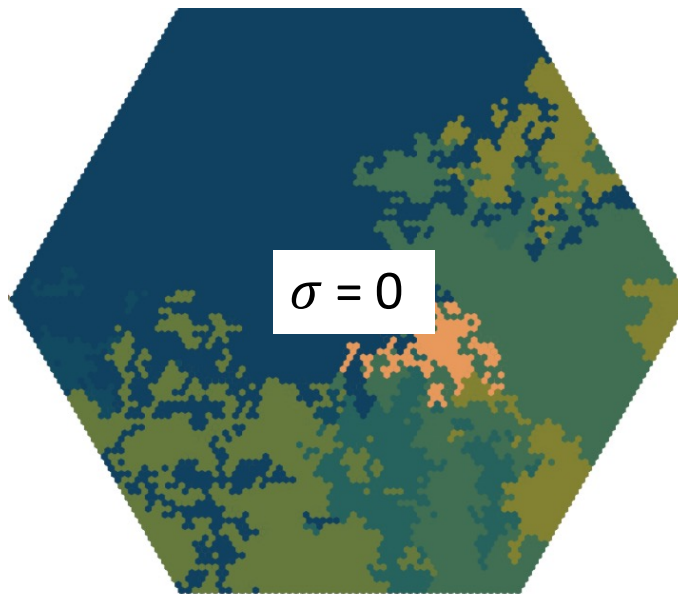
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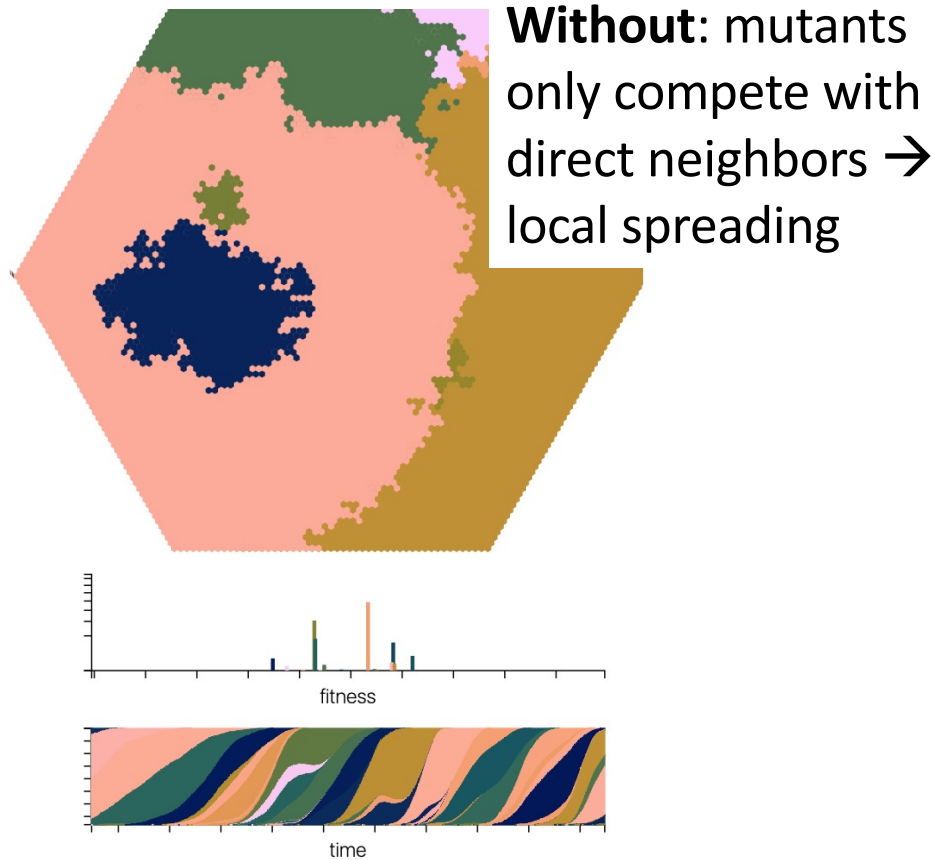


Parameters: Selectivity σ

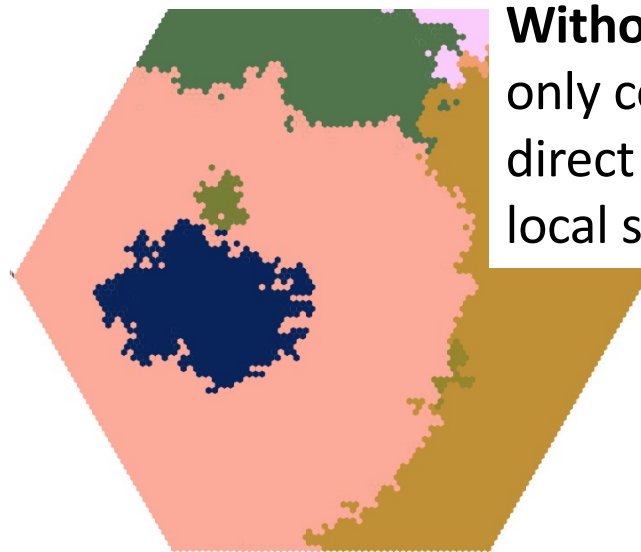
= controls how clear decision about winner is



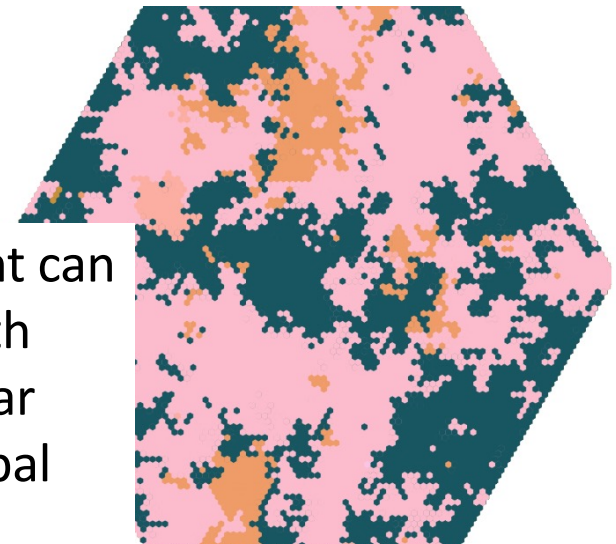
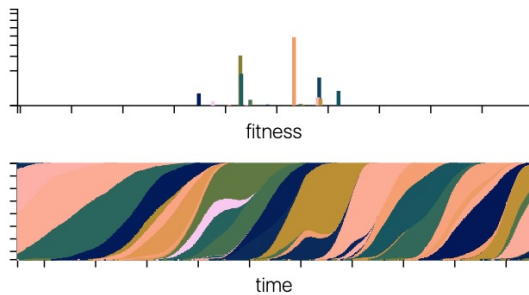
Parameters: long range interactions



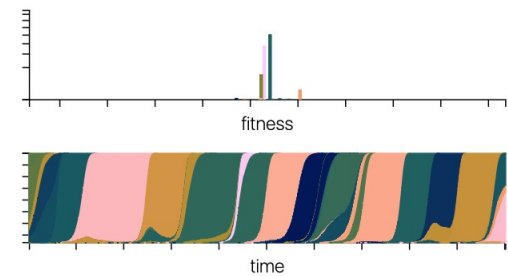
Parameters: long range interactions



Without: mutants only compete with direct neighbors → local spreading



With: mutant can compete with individuals far away → global spreading



Limitations

Limitations

not the same probability of having beneficial or non-beneficial mutation (mostly neutral mutations)

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not the same probability of having beneficial or non-beneficial mutation (mostly neutral mutations)

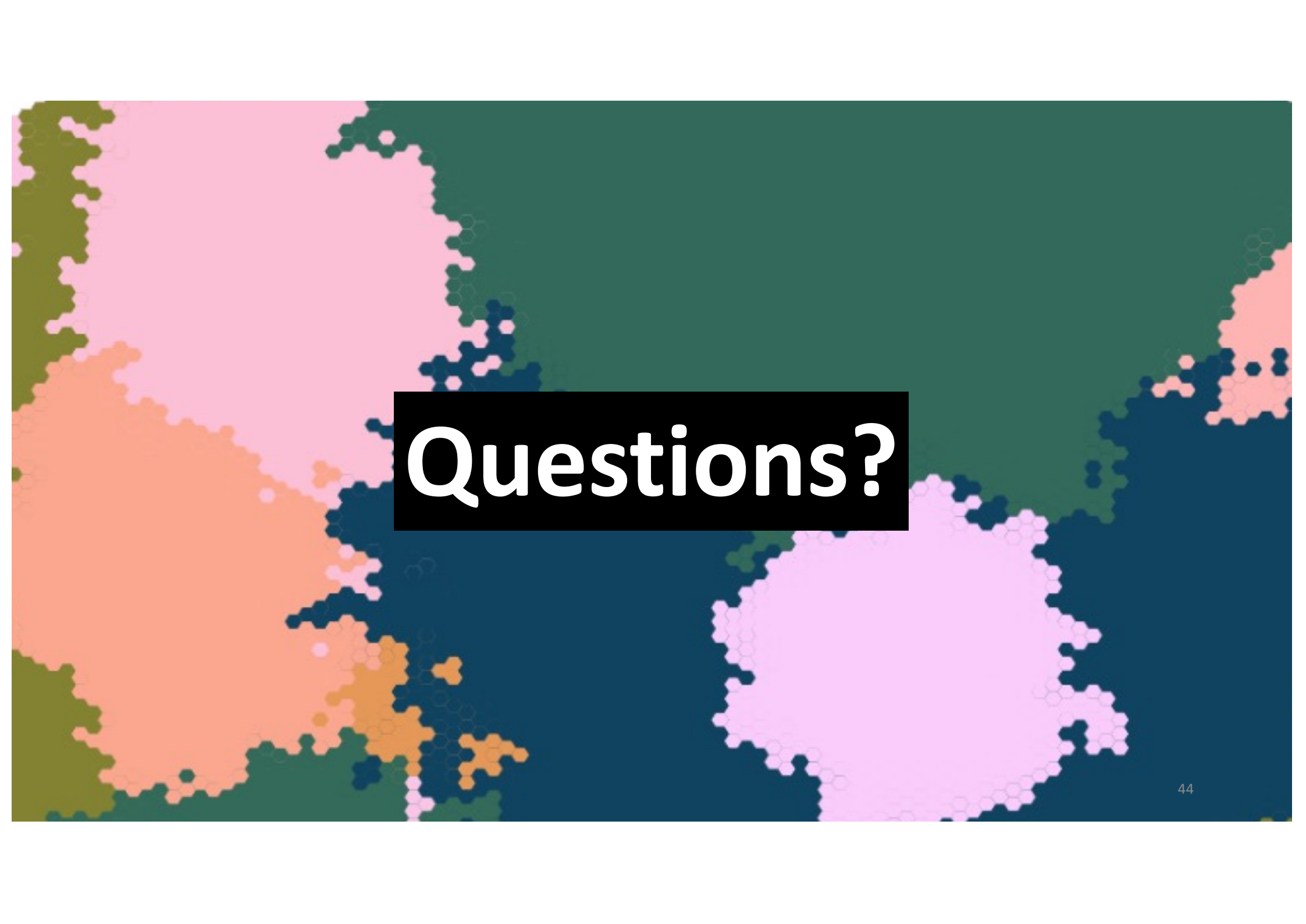
Simplification of fitness

Limitations

not the same probability of having beneficial or non-beneficial mutation (mostly neutral mutations)

Simplification of fitness

Homogenous population + remains constant



Questions?

Sources

- <https://www.complexity-explorables.org/explorables/a-patchwork-darwinge/>
- <https://www.geogebra.org/calculator>
- Lectures „Evolution und Biodiversität“ by Oliver Otti
- Champagnat, N., Ferrière, R., & Méléard, S. (2006). Unifying evolutionary dynamics: From individual stochastic processes to macroscopic models. *Theoretical Population Biology*, 69(3), 297–321.
<https://doi.org/10.1016/j.tpb.2005.10.004>

Images

- <https://www.historyhit.com/culture/1859-darwins-revolutionary-origin-species-published/>
- <https://de.pinterest.com/pin/funny-elephant-gifs--674977062895773866/>